

Time & Work — Practice Questions

Medium to Difficult, With Full Solutions

Aptitude Grind

Difficulty is marked as [M] Medium or [H] Hard. Solve on your own first, then check the solution. Solutions use the methods from `01-fundamentals.tex`, `02-fast-tricks-and-shortcuts.tex`, and `03-question-pattern-catalog.tex`. Every numeric answer below was cross-checked independently using exact fraction arithmetic (Python's `fractions.Fraction`) before being finalized.

Q1 [M] — Two People Working Together (Pattern A1)

A can complete a job alone in 12 days, B alone in 24 days. In how many days can they finish it working together?

Solution

$$\text{Together} = \frac{12 \times 24}{12 + 24} = \frac{288}{36} = 8$$

Answer: 8 days

Q2 [M] — Efficiency Ratio (Pattern B1)

The efficiency ratio of A to B is 3:5. A alone takes 20 days to complete a job. Find the number of days B alone would take, and the number of days they would take working together.

Solution

Efficiency ratio 3:5 means A is the *less* efficient one, so A (efficiency 3) takes more time than B (efficiency 5). Since time is inversely proportional to efficiency:

Time ratio $A : B = 5 : 3$ (reverse of efficiency ratio)

$$A = 20 \text{ days} \Rightarrow B = 20 \times 3/5 = 12 \text{ days}$$

$$\text{Together} = \frac{20 \times 12}{20 + 12} = \frac{240}{32} = 7.5 \text{ days}$$

Answer: B alone = 12 days; together = 7.5 days

Q3 [M] — The Difference Formula (Pattern §5 of 01-fundamentals.tex)

A alone can complete a job in 30 days. A and B together can complete it in 12 days. In how many days can B alone complete the job?

Solution

$$\frac{1}{B} = \frac{1}{12} - \frac{1}{30} = \frac{5}{60} - \frac{2}{60} = \frac{3}{60} = \frac{1}{20}$$

Answer: B alone = 20 days

Q4 [M] — LCM Method, Three Workers (Pattern A2)

A, B, and C can individually complete a job in 12, 15, and 20 days respectively. In how many days will they finish it working together?

Solution

LCM(12, 15, 20) = 60 units of total work

A's rate = $60/12 = 5$, B's rate = $60/15 = 4$, C's rate = $60/20 = 3$ (units/day)

Combined rate = $5 + 4 + 3 = 12$ units/day

Time = $60/12 = 5$ days

Answer: 5 days

Q5 [M] — Pipes Filling and Emptying Together (Pattern E1)

A pipe can fill a tank in 6 hours; another pipe can empty the full tank in 9 hours. If both pipes are opened together, in how much time will the tank be filled?

Solution

$$\text{Net rate} = \frac{1}{6} - \frac{1}{9} = \frac{3}{18} - \frac{2}{18} = \frac{1}{18} \Rightarrow \text{Time} = 18 \text{ hours}$$

Answer: 18 hours

Q6 [H] — Pipe Closed Partway Through (Pattern E2)

Fill pipe A can fill a tank in 10 hours; outlet pipe B can empty it in 15 hours. Both are opened together, but B is closed after 3 hours. Find the total time to fill the tank.

Solution

Net rate (both open) = $1/10 - 1/15 = 1/30$

Work done in first 3 hours = $3 \times 1/30 = 1/10$

Remaining work = $1 - 1/10 = 9/10$

A alone (rate $1/10/\text{hr}$) needs: $(9/10) \div (1/10) = 9$ more hours

Total time = $3 + 9 = 12$ hours

Answer: 12 hours

Why This Is a Common Trap

People sometimes keep using the *net* rate for the whole problem, forgetting that once the outlet closes, only the inlet's rate applies for the rest of the time. Always split into phases the moment any pipe's status changes.

Q7 [H] — Worker Joining Midway (Pattern C1)

A can complete a job alone in 20 days. A works alone for 5 days, then B (who alone would take 30 days) joins. How many total days does the job take to finish?

Solution

$$\text{Work done by A in 5 days} = 5/20 = 1/4$$

$$\text{Remaining work} = 3/4$$

$$\text{Combined rate (A + B)} = 1/20 + 1/30 = 3/60 + 2/60 = 1/12$$

$$\text{Extra days needed} = (3/4) \div (1/12) = 9$$

$$\text{Total days} = 5 + 9 = 14$$

Answer: 14 days

Q8 [H] — Workers Leaving Midway (Pattern C2)

36 workers can complete a job in 20 days. After 5 days, 6 workers leave the job. In how many more days will the remaining workers finish it?

Solution

$$\text{Total work} = 36 \times 20 = 720 \text{ worker-days}$$

$$\text{Work done in 5 days} = 36 \times 5 = 180 \text{ worker-days}$$

$$\text{Remaining work} = 720 - 180 = 540 \text{ worker-days}$$

$$\text{Workers remaining} = 36 - 6 = 30$$

$$\text{Days needed} = 540/30 = 18$$

Answer: 18 more days

Q9 [M] — Alternate-Day Working (Pattern F1)

A can complete a job alone in 6 days, B alone in 9 days. Working on alternate days, starting with A, in how many days will the job be finished?

Solution

$$\text{2-day cycle work} = \frac{1}{6} + \frac{1}{9} = \frac{3}{18} + \frac{2}{18} = \frac{5}{18}$$

Three full cycles (6 days) complete $3 \times 5/18 = 15/18 = 5/6$ of the work. Remaining work = $1/6$. Day 7 is A's turn again: A's rate is $1/6$, so A finishes exactly the remaining $1/6$ in **one more full day**.

$$\text{Total} = 6 \text{ (three full cycles)} + 1 \text{ (A's final day)} = 7 \text{ days}$$

Answer: 7 days

Q10 [H] — Men/Women Equivalence (Pattern D1)

3 men do the same amount of work as 5 women (equal efficiency comparison). 3 men alone can complete a job in 20 days. In how many days can 6 men and 10 women together complete the same job?

Solution

$$3 \text{ men alone: } 20 \text{ days} \Rightarrow 1 \text{ man's rate} = 1/(3 \times 20) = 1/60$$

$$3 \text{ men} = 5 \text{ women (equal output)} \Rightarrow 1 \text{ woman's rate} = (3 \times 1/60)/5 = 1/100$$

$$6 \text{ men} + 10 \text{ women rate} = 6 \times (1/60) + 10 \times (1/100) = 1/10 + 1/10 = 1/5$$

$$\text{Time} = 5 \text{ days}$$

Answer: 5 days

Note

6 men and 10 women is exactly *double* the “3 men = 5 women” equivalence group, so it makes sense the answer is a clean number — doubling the workforce on a fixed job always halves the time, and here the group ratio was scaled up perfectly.

Q11 [M] — Wages Based on Work (Pattern G1)

A can complete a job alone in 8 days, B alone in 12 days. They work together and are jointly paid Rs. 2500. Find each person's share.

Solution

$$\text{Rate ratio } A : B = 1/8 : 1/12 = 3 : 2 \quad (\text{multiply both by } 24)$$

$$A's \text{ share} = 2500 \times 3/5 = \text{Rs. } 1500$$

$$B's \text{ share} = 2500 \times 2/5 = \text{Rs. } 1000$$

Answer: A = Rs. 1500, B = Rs. 1000

Q12 [M] — Efficiency Decrease → Time Change (Pattern J2)

A's efficiency decreases by 25%. If A originally took 18 days to complete a job at full efficiency, how many days will A now take?

Solution

$$\text{New Time} = \text{Old Time} \times \frac{100}{100 - 25} = 18 \times \frac{100}{75} = 24$$

Answer: 24 days

Q13 [H] — Variable Efficiency Mid-Job (Pattern J1)

Working at a constant rate, A can finish a job in 12 days. A works at the normal rate for the first 3 days, then works at **triple** efficiency for the rest of the job. How many total days does the job take?

Solution

$$\text{Normal rate} = 1/12$$

$$\text{Work done in first 3 days} = 3 \times 1/12 = 1/4$$

$$\text{Remaining work} = 3/4$$

$$\text{Tripled rate} = 3 \times 1/12 = 1/4$$

$$\text{Extra days needed} = (3/4) \div (1/4) = 3$$

$$\text{Total days} = 3 + 3 = 6$$

Answer: 6 days

Q14 [H] — Percentage of Work Completed (Pattern I1)

A can complete 90% of a job in 18 days. In how many days can A alone complete the entire job? B then finishes the remaining 10% of the job in 3 days — in how many days could B alone complete the entire job?

Solution

$$A : 90\% \text{ in } 18 \text{ days} \Rightarrow 100\% \text{ in } 18/0.9 = 20 \text{ days}$$

$$B : 10\% \text{ in } 3 \text{ days} \Rightarrow 100\% \text{ in } 3/0.1 = 30 \text{ days}$$

Answer: A alone = 20 days; B alone = 30 days

Why This Is a Common Trap

People sometimes try to combine A and B's *individual* full-job times directly (e.g., averaging 20 and 30) to answer a different question about their combined time. Here the question

only asks for each person's **individual** full-job time — scale each person's given fraction independently; don't mix the two workers' numbers together unless the question specifically asks for a combined rate.

Answer Key Summary

Answers

Question	Answer
Q1	8 days
Q2	B = 12 days; together = 7.5 days
Q3	20 days
Q4	5 days
Q5	18 hours
Q6	12 hours
Q7	14 days
Q8	18 more days
Q9	7 days
Q10	5 days
Q11	A = Rs. 1500, B = Rs. 1000
Q12	24 days
Q13	6 days
Q14	A = 20 days, B = 30 days